



MECH MAINT- OFF SITE Department
SIMHADRI

Ref:SMTTP/MECH MAINT-/2022-23/NTPCSIMH/17845

Date:

07/04/2022(16:41:16)

Subject:APPROVAL OF LMI- RELIABILITY IMPROVEMENT MATRIX
FOR COOLING TOWER

For monitoring the performance and condition of cooling towers and improve their reliability at various stations, a common Reliability Improvement Matrix has been developed by Corporate OS as Operation Information Notes (OIN) in April 2022 vide document no. COS-ISO-00- OIN/OPS/SYST/081 (copy enclosed). Based on the OIN, a draft LMI has been prepared for Simhadri plant for performance and condition monitoring of NDCTs of main plant.

The draft LMI 'Reliability Improvement Matrix of Cooling Towers' is hereby submitted for review of all concerned and thereafter, approval of competent authority.

Submitted by:

जे.पी. सिंह - वरिष्ठ प्रबंधक

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On:07/04/2022(16:41:16) **Ref:**SMTTP/MECH MAINT-/2022-23/NTPCSIMH/17845

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May be approved.

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Review as discussed and define the role of EMD, if any

Submitted by:

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Review the document as per the note of EMD .

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On:16/04/2022(06:27:06) **Ref:**SMTTPP/MECH MAINT-/2022-23/NTPCSIMH/17845

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On:16/04/2022(15:37:28) **Ref:**SMTPP/MECH MAINT-/2022-23/NTPCSIMH/17845

Reviewed

Submitted by:

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On:19/04/2022(19:12:57) **Ref:**SMTPP/MECH MAINT-/2022-23/NTPCSIMH/17845

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Submitted for approval of competent authority

Submitted by:

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Please modify the page no.3 of the proposed LMI, based on current positions held.

Also please specify after taking clarification from COS as to the periodicity of the Reliability Improvement Matrix evaluation of the Cooling towers in the proposed LMI, i.e. at what frequency this evaluation (as per clause no. 6.0) of the cooling Tower is to be done.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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May please be reviewed, as discussed.

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Sir,

With reference to the comments mentioned in prepages, the same was discussed with Sr. Mgr. (CC-OS, Raipur). Accordingly, the LMI has been updated. Revised LMI file (APPROVED_ LMI_Reliability Improvement Matrix of Cooling Towers R1) is attached with Note.

Submitted by:

जे.पी. सिंह - वरिष्ठ प्रबंधक

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May please be approved

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On:05/05/2022(10:55:31) **Ref:**SMTTPP/MECH MAINT-/2022-23/NTPCSIMH/17845

Operation side points noted and forwarded for further approval

Submitted by:

राजशेखर पाला - अपर महाप्रबंधक

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May please be accorded approval by HOP.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:09/05/2022(10:10:22) **Ref:**SMTTPP/MECH MAINT-/2022-23/NTPCSIMH/17845

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

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Approved & Submitted by:

गिरीश चन्द्र चौकसे - महाप्रबंधक

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On:09/05/2022(17:37:54) **Ref:**SMTPP/MECH MAINT-/2022-23/NTPCSIMH/17845

NTPC LTD.
SIMHADRI-COAL

TECHNICAL COMPLIANCE DOCUMENT
LOCATION MANAGEMENT INSTRUCTION

LMI/SMTTPP/OFS-08/OIN/OPS/SYST/081

Issue no. 1, Rev. No.: 0 Date: APRIL 2022

Reliability Improvement Matrix for
Cooling Tower

Approved for Implementation by: HOP, NTPC SIMHADRI

LMI Name	Reliability Improvement Matrix of Cooling Tower	SIMHADRI
LMI No.	LMI/SMTTP/OFS-08/OIN/OPS/SYST/081	

LMI Name	Reliability Improvement Matrix of Cooling Tower	
LMI No.	LMI/SMTTP/OFS-08/OIN/OPS/SYST/081	
Reference Document	COS-ISO-00- OIN/OPS/SYST/081	
Prepared by	Name:	JP Singh
	Designation:	Sr. Manager (Offsite MM)
Checked by	Name:	D Sri Rama Sastry
	Designation:	DGM (offsite MM)
Reviewed by	Name:	Sh. Jayanthi Gopi Kiran
	Designation:	DGM (Civil)
	Name:	A.K.Mukhopdhyay
	Designation:	AGM (offsite MM & CIVIL)
	Name:	Sh. P .R .Sekhar
	Designation:	AGM (Operation)
	Name:	Sh. N.Venugopal
	Designation:	AGM (Chemistry)
	Name:	Sh. G.S.Fanikiran
	Designation:	DGM (MTP)
	Name:	Ms. Jhimli Dhar
	Designation:	AGM (EEMG)
	Name:	Sh. M.E.Kovoor
	Designation:	GM (Maint)
	Name:	Sh. R K Jha
	Designation:	GM (O&M)
Approved for issue	Name:	Girish Chandra Choukse
	Designation:	HoP, SMTTP- Simhadri

LMI Name	Reliability Improvement Matrix of Cooling Tower	SIMHADRI
LMI No.	LMI/SMTTP/OFS-08/OIN/OPS/SYST/081	

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LMI Name	Reliability Improvement Matrix of Cooling Tower	SIMHADRI
LMI No.	LMI/SMTTP/OFS-08/OIN/OPS/SYST/081	

RELIABILITY IMPROVEMENT MATRIX OF COOLING TOWER

1.0 INTRODUCTION

Cooling Towers are one of the important parts of Power Plants. The purpose of A Cooling Tower is to reject the process heat absorbed by the cooling water to the atmosphere. The hot water from the process is conveyed through the water distribution system and is uniformly distributed over the cooling fill arranged inside the cooling tower where it is cooled by direct contact with the cooling air stream passing in a counter flow/ cross flow direction. The required airflow is supplied by induced draft fans in case of IDCT and naturally due to density difference of cold outside air and hot humid air inside in case of NDCT. The heat transfer is affected by evaporation & convection.

To improve the reliability of Cooling Towers it is envisaged to formulate a reliability improvement framework. Consequently, CT Reliability Improvement Matrix has been prepared, the compliance of which shall be used as basis for evaluation of performance of CT for all stations.

2.0 SCOPE

CT Reliability Improvement Matrix will help in improving the reliability of CT equipment and process of cooling. The compliance of this matrix shall be used as basis for evaluation of performance of CT for Simhadri Thermal plant. Accordingly, weightages have been allocated to respective parameter for evaluation. It will also enable internal benchmarking among all the CT's of the station. CT Reliability Improvement Matrix will be evaluated by respective Regional OS (ROS) during pre-ORT visit every month. The evaluation shall be shared with COS by ROS.

All main NDCTs of thermal plant will be under the scope of evaluation.

3.0 REFERENCE DOCUMENT

COS-ISO-00-OIN/OPS/SYST/081

4.0 REVISION HISTORY

Nil

5.0 SUPERSEDED DOCUMENTS

Nil

LMI Name	Reliability Improvement Matrix of Cooling Tower	SIMHADRI
LMI No.	LMI/SMTTP/OFS-08/OIN/OPS/SYST/081	

6.0 RELIABILITY IMPROVEMENT MATRIX OF CT SYSTEM

Station:		Region:	Year:	Month:		
S. No.	Criteria	Criteria Description	Reference Document	Evaluation Criteria	Weightage	Rating
1.	Performance Indicators	i Generation Loss due to CT	SAP Report by EEMG	If no Loss- 10. up to 0.1 MU/day- 5, >0.1 MU/day- 0	10.0	
		ii CT Effectiveness	EEMG Report vs Design	If Deviation 0-10% -5, Deviation 10-20% - 2.5, Otherwise- 0	5.0	
		iii CT Capability	EEMG Report	Above 90%- 10; >85-90%- 7; >80-85%- 5; >75-80%- 3; < 75 %- 0	10.0	
2.	System Compliance	i. Maintenance Performance Index	SAP Report by MTP: Completed vs Raised	>90- 100%- 3; >80-90%- 2; >70- 80 %- 1; < 70%- 0	3.0	
3.	Open Approach Channel (Prevention of Cooling Water Contamination)	i. Installation of double stage bucket strainer at CT outlet channel	Joint surveillance checks with Offsite, Operation & Civil Maint. Dept. by ORT Coordinator.	If found Good- 5; Satisfactory- 3; else- 0	5.0	
		ii. Parapet wall construction on both sides of OAC to prevent ingress of ash/ mud/ debris and rainwater		If found Good- 2; satisfactory- 1; else- 0	2.0	
		iii. Provision of 2 m (approx.) pavement on either side of OAC to ensure area near the channel is free from vegetation		If found Good- 2; satisfactory- 1; else- 0	2.0	
		iv. Installation of Tray beneath the ash/sludge pipelines crossing OAC		Available in Healthy Condition- 1, otherwise-0	1.0	
		v. Construction of Parapet Wall Along the Ash Pipelines passing near CT Pipelines		Available in Healthy Condition- 1, Otherwise-0	1.0	

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Station:		Region:	Year:		Month:	
S. No.	Criteria	Criteria Description	Reference Document	Evaluation Criteria	Weightage	Rating
		vi. Covering of OAC with durable Nylon net to prevent fall of debris into the channel		Available in Healthy Condition- 1, otherwise-0	1.0	
		vii. OAC Walls to be Clear of any growth of vegetation		No Vegetation- 1, else- 0	1.0	
4.	Cooling Tower (Indicators of CT Performance) & Support Infrastructures	i. Water flow pattern across cells	Joint Protocol by Operation and Offsite Reps	Uniform Flow in all the cells- 2, > 50% Cells- 1, <50% cells- 0	2.0	
		ii. Nozzles Healthiness	Joint Protocol by Operation and Offsite Reps	If Complied- 2, or else- 0	2.0	
		iii. Healthiness of Drift eliminators	Joint Protocol by Operation and Offsite Reps	If 100% complied- 2, otherwise- 0	2.0	
		iv. Replacement of damaged fills	Replacement Record by Offsite	If 100% complied- 2, otherwise- 0	2.0	
		v. CT Cold Basin Sludge Pumps healthiness	Offsite & Operation Inspection Report	If Working Effectively- 1, otherwise- 0	1.0	
		vi. CT area pavement/ Vegetation Removal/ Parapet Wall: 30 m. area (approx.) on either side of CT (Air Flow Direction) to be free of Vegetation and properly Paved to ensure free air flow path. Construction of Parapet Wall around CT cold basin area.	Joint surveillance check by Civil, Offsite and ROS reps	Pavement Done-1, else-0. No Vegetation- 1 else- 0. Parapet Wall- 1, Else- 0	3.0	
5.	Chemistry Parameters: Turbidity	i. To be maintained as per OGN	Chemistry Report	If less than Recommendation- 1, Else- 0	1.0	

LMI Name	Reliability Improvement Matrix of Cooling Tower	SIMHADRI
LMI No.	LMI/SMTTP/OFS-08/OIN/OPS/SYST/081	

Station:		Region:	Year:	Month:		
S. No.	Criteria	Criteria Description	Reference Document	Evaluation Criteria	Weightage	Rating
	Algae and scale formation in CT internals	ii. Effective Chemical Dozing/ To be removed/ cleaned periodically	Joint surveillance checks by Offsite and ROS Reps.	No Algae or Scaling- 2, else- 0	2.0	
6.	Cleanliness, House Keeping, Environment & Safe Working Conditions	i. Safe approach to the CT area, Hot water duct and CT internals area	Joint surveillance checks by Offsite and ROS Reps.	If overall rating by ROS is Good- 4, If Satisfactory then- 2, if Poor then- 0.	4.0	
		ii. Lighting/ Illumination				
		iii. Safe working conditions including handrail, toe guards, sealing of floor openings, Barricade for any opening etc.				
7.	Critical spares availability	i. Availability of recommended spares for partial loss prevention	Offsite Department report	100% all items availability- 4, 80% items available-2.5, less than 80% items available- 0.	4.0	
8.	Safety Practices	i. Use of LOTO Kit for Mechanical Isolation.	Offsite Department report	If 100% complied- 2.5, otherwise- 0	2.5	
		ii. JSA	Recent JSA Copy linked with PTW	If 100% complied- 2.5, otherwise- 0	2.5	
		iii. Load Testing of Lifting Tools & Tackles, Hoists & Lifts as per schedule	Offsite Department report/ Safety report	If 100% complied- 2.5, otherwise- 0	2.5	
		iv. Availability & Use of Safety Net along with Safety Kit (e.g., full body harness, lifeline, rope grab, fall arrestor) for working at height and deep duct	Offsite Department report	If 100% complied- 2.5, otherwise- 0	2.5	
		v. Lightening arrestors connection healthiness for CT	Joint Protocol by Operation and Offsite Reps	If 100% complied- 2, otherwise- 0	2.0	
		vi. Aviation lights for NDCTs	Joint Protocol by Operation and Offsite Reps	If 100% complied- 1, otherwise- 0	1.0	

LMI Name	Reliability Improvement Matrix of Cooling Tower	SIMHADRI
LMI No.	LMI/SMTTP/OFS-08/OIN/OPS/SYST/081	

Station:		Region:	Year:	Month:		
S. No.	Criteria	Criteria Description	Reference Document	Evaluation Criteria	Weightage	Rating
9.	Structure check, Painting and aesthetic	i. CT Civil Structure inspection & Rectification Fixture Healthiness of Monorail inside Stack/Lightening Arrester Any Reinforcement Exposure	Joint protocol by Civil and Offsite Reps during overhaul	Structures & Fixtures: if restoration done- 2, If detailed investigation done- 1, otherwise -0	2.0	
		ii. Painting of Riser pipes & above ground ducts	Rolling Plan	Rating by ROS is satisfactory- 2, if not- 0.	2.0	
10.	New Technology/ New Initiatives/ PIP	i. Identified and defined by site, site specific. At least two no. to be identified.	Offsite Department Report	100% implementation-2, 50% implementation- 1, otherwise -0	2.0	

Note 1: For stations not having specific equipment/practices not applicable to station mentioned in above evaluation Criteria, rating will be done as per actual basis and extrapolated to 100 Marks.

Note 2: SAP report for generation loss and PM compliance, Chemistry report, recent PTW with JSA has been verified and also other reports for example Service reports/ Protocols/ COS formats/ Logbook records verified/ randomly verified.

LMI Name	Reliability Improvement Matrix of Cooling Tower	SIMHADRI
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6.1 THE REPORT SHOULD BE DISTRIBUTED TO THE FOLLOWING:

- a. HOD (EEMG)
- b. HOD (MTP)
- c. HOD (Chemistry)
- d. HOD (Operation)
- e. GM (Maint.)
- f. GM (O&M)
- g. HOP, Simhadri
- h. GM (OS-SR-HQ) - Respective Region

7.0 RESPONSIBILITY CENTER FOR IMPLEMENTATION

Overall responsibility for implementation of the LMI will be with Offsite MM.

8.0 EVALUATION MATRIX

The evaluation sheet as in 6.0 above shall be filled based on the reference documents submitted by the departments and evaluation by ROS and will be submitted by Offsite MM.

9.0 THE EXCEPTIONS/ DEVIATIONS

In the OIN, some points included in evaluation sheet for IDCTs have been omitted as simhadri-Thermal plant doesn't have any IDCT.

10.0 RECORD KEEPING

Responsibility for generating, sharing with Offsite MM and keeping the record for fulfilment of the CT evaluation matrix will be as per the table below:

Responsible Departments	S. No. of Evaluation Criteria
OFFSITE MM	3. (i., ii., iii., iv., v., vi. & vii.), 4. (i., ii., iii. & v.), 5. (ii.), 6. (i., ii. & iii.), 7. (i.), 8. (i., ii., iii. & iv.), 9. (ii.), 10. (i.)
EEMG	1. (i., ii. & iii.)
MTP	2. (i.)
CHEMISTRY	5. (i)
OPERATION	4. (iv.), 8. (v. & vi.)
CIVIL	9. (i.)

11.0 REVIEWER

Offsite MM will be responsible for initiating modification proposals in this document.

12.0 REVIEW

Review of this document is to be done every two years or as per the site requirements or any changes in the OIN.

LMI Name	Reliability Improvement Matrix of Cooling Tower	SIMHADRI
LMI No.	LMI/SMTTP/OFS-08/OIN/OPS/SYST/081	

13.0 APPROVING AUTHORITY

HOP Simhadri is the approving authority for this LMI.